

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 96-141
WASTE DISCHARGE REQUIREMENTS
FOR
PACIFIC GAS AND ELECTRIC COMPANY
HELMS PUMPED STORAGE PROJECT
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Pacific Gas and Electric Company (hereafter Discharger), a California corporation, owns and operates a wastewater treatment facility (WWTF) serving its Helms Pumped Storage Project headquarters and company homes near the headquarters at Wishon and Courtright reservoirs on the Kings River drainage. The WWTF consists of an extended aeration package treatment plant and a spray disposal field (hereafter "sprayfield") for final disposal.
2. Waste Discharge Requirements Order No. 96-017, adopted by the Board on 26 January 1996, prescribes secondary effluent limits and disinfection requirements for a year-round discharge of up to 15,000 gallons per day (gpd) from the headquarters package plant to the sprayfield.
3. Order No. 96-017 does not reflect the actual treatment capacity of the WWTF.
4. The Discharger submitted a technical report dated 15 February 1996, which indicates that the package plant has treatment capacity of only 7,500 gpd.
5. The WWTF serves six cabins owned by the Discharger and used for housing of about 16 people involved in maintenance, operation, and management of the Discharger's hydroelectric power facilities. The waste discharge from cabins is 2,000 to 4,000 gpd year-round. The remaining flow is from the headquarters building, a 14-bed dormitory, a 24-bed dormitory, a kitchen facility, recreation hall, guest house, washrack/garage facility, and inflow/infiltration. Up until mid-1995, the WWTF also treated domestic wastes from a privately owned campground and associated inflow/infiltration. The Discharger disconnected the campground from its WWTF to reduce influent flows. The campground is discharging to its own septic tank/leachfield system.
6. Sludge from the WWTF is disposed of off-site by a licensed septic tank pumper. In recent years, the plant has been hydraulically overloaded due to excessive precipitation that may get into the collection system in the form of inflow/infiltration via cleanouts and broken

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pipes. In June 1995, the plant was also organically overloaded because solids from unit processes had not been removed regularly. Monthly average flows consistently exceed the 15,000 gpd rated plant capacity. The Discharger proposes to identify the source of inflow/infiltration and take the necessary steps to minimize excessive inflow/infiltration.

7. The WWTF and its sprayfield are about 7,000 feet above Mean Sea Level in Section 1, T11S, R27E, MDB&M, with surface water drainage to the North Fork of the Kings River, as shown in Attachment A, which is attached hereto and part of this Order by reference. The site lies within the North Fork Kings River hydrologic subarea (No. 552.33), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
8. Average annual precipitation for the site is about 50 inches, ninety percent of which is snow. The sprayfield is surrounded by surface water drainage channels and on a hillside with slopes in excess of 40 percent. During the winter months, the sprayfield is not easily accessible because it is covered with snow for up to six months. Snowmelt enters the Kings River and other ephemeral streams that drain to the Valley floor. Under these circumstances, disinfection limits and effective runoff control measures are essential.
9. The area is characterized by numerous springs and ephemeral streams, which form right after snow melts. Soils consist of a couple of inches of humus accumulations, but they are mainly granite, decomposed granite, and boulders. Areal ground water is typically four to ten feet below ground surface and of excellent quality.
10. The Board adopted the *Water Quality Control Plan for the Tulare Lake Basin, Second Edition*, (hereafter Basin Plan), which designates beneficial uses and contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
11. The beneficial uses of Wishon Reservoir identified in the Basin Plan are water contact recreation; noncontact water recreation; fresh water replenishment; hydropower generation; preservation and enhancement of fish, wildlife, and other aquatic resources; and preservation of rare and endangered species.
12. The beneficial uses of underlying ground water identified in the Basin Plan are domestic, industrial, and agricultural supply.
13. The permitted discharge is consistent with the antidegradation provisions of State Water Resources Control Board Resolution 68-16. Continued discharge to the sprayfield will not cause significant impacts on ground water and surface waters and depletion of limited ground water resources. The discharge allows wastewater services essential for the reliable

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generation of hydroelectric power for the San Joaquin Valley, which accommodates housing and economic expansion in the Valley, and is considered to be a benefit to the people of the State. The Discharger proposes to use best practicable treatment methods to comply with Resolution 68-16.

14. The action to revise waste discharge requirements for this existing permitted facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
15. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
16. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 96-017 is rescinded and the Pacific Gas and Electric Company, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, California Code of Regulations (CCR), Section 2510, et seq. (hereafter Chapter 15), is prohibited.
4. The direct or indirect discharge of wastes outside the sprayfield is prohibited.

B. Discharge Specifications

1. The monthly average discharge flow shall not exceed 7,500 gpd.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

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3. Wastewater within the aeration chamber of the treatment plant shall have a dissolved oxygen concentration of at least 2.0 mg/l at all times.
4. Effluent from the treatment plant shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Monthly Median</u>	<u>Daily Maximum</u>
Settleable Solids	ml/l	0.5		1.0
BOD	mg/l	30		60
Suspended Solids	mg/l	30		60
Total Coliform Organisms	MPN/100 ml		23	240

5. The discharge to the sprayfield shall be managed to minimize erosion and runoff from the disposal area.
6. The maximum specific electrical conductivity (EC) of the discharge shall not exceed 300 μ mhos/cm over the EC of the source water or a maximum of 700 μ mhos/cm, whichever is less.
7. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.

C. Sludge Disposal

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.

D. Ground Water Limitations

The discharge, in combination with other sources, shall not cause underlying ground water to contain waste constituents in concentrations statistically greater than background water quality.

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E. Provisions

1. The Discharger shall comply with Monitoring and Reporting Program No. 96-141, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

4. The Discharger shall use the best practicable control technique currently available to comply with this Order.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, ground water, cooling waters, and condensates that are essentially free of pollutants.

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7. The Discharger shall submit to the Board on or before each report due date the specified document or, if an action is specified, a written report detailing evidence of compliance with the date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when it returns to compliance with the time schedule.
8. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
9. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes." Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.
10. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 3 May 1996.


WILLIAM H. CROOKS, Executive Officer

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CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 96-141
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Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to Board and attached to this Program.

INFLUENT MONITORING

Influent samples shall be collected at the inlet of the headworks and approximately the same time as effluent samples. Influent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Settleable Solids	ml/l	Grab	Weekly
BOD ₅ ¹	mg/l	Grab	2/month
Total Suspended Solids	mg/l	Grab	2/month

¹ Five-day, 20° Celsius biochemical oxygen demand.

EFFLUENT MONITORING

Except for flow, which may also be measured at the headworks, effluent samples shall be collected prior to discharge to the sprayfield. Effluent samples should be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency¹</u>
Total Daily Flow	gpd	Continuous	Daily
Settleable Solids	ml/l	Grab	Weekly
Electrical Conductivity @ 25°C	µmhos/cm	Grab	Weekly
BOD ₅ ²	mg/l	Grab	2/month

¹ (See footnotes next page).

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<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency¹</u>
Total Suspended Solids	mg/l	Grab	2/month
Total Coliform Organisms	MPN/100 ml	Grab	2/month

- ¹ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly average for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.
- ² Five-day, 20° Celsius biochemical oxygen demand.

SPRAYFIELD MONITORING

The Discharger shall inspect the sprayfield weekly from November to April and monthly the rest of the year. In conducting the inspection, a log shall be kept of the condition of the sprayfield. Special attention shall be given to the presence or absence of vectors, objectional odors, erosion, and effluent runoff. A summary, addressing the conditions of the sprayfield, shall be included with the monthly monitoring report. Where remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken. When hazardous conditions prevent the Discharger from conducting the inspections at the specified frequency, the Discharger shall note the conditions in the log and specify in the monthly monitoring report the alternative monitoring measures it is taking to assure compliance with Discharge Specification No. B.5.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Electrical Conductivity @ 25°C	µmhos/cm	Annually ¹

- ¹ If the source water is from more than one well, the EC shall be reported as a weighted average and include copies of supporting calculations.

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REPORTING

Monthly monitoring reports shall be submitted to the Board by the **20th day** of the following month.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **31 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, titles, certificate grade (if any) and expiration date, and general responsibilities of persons operating and maintaining the wastewater treatment plant.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).
- d. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

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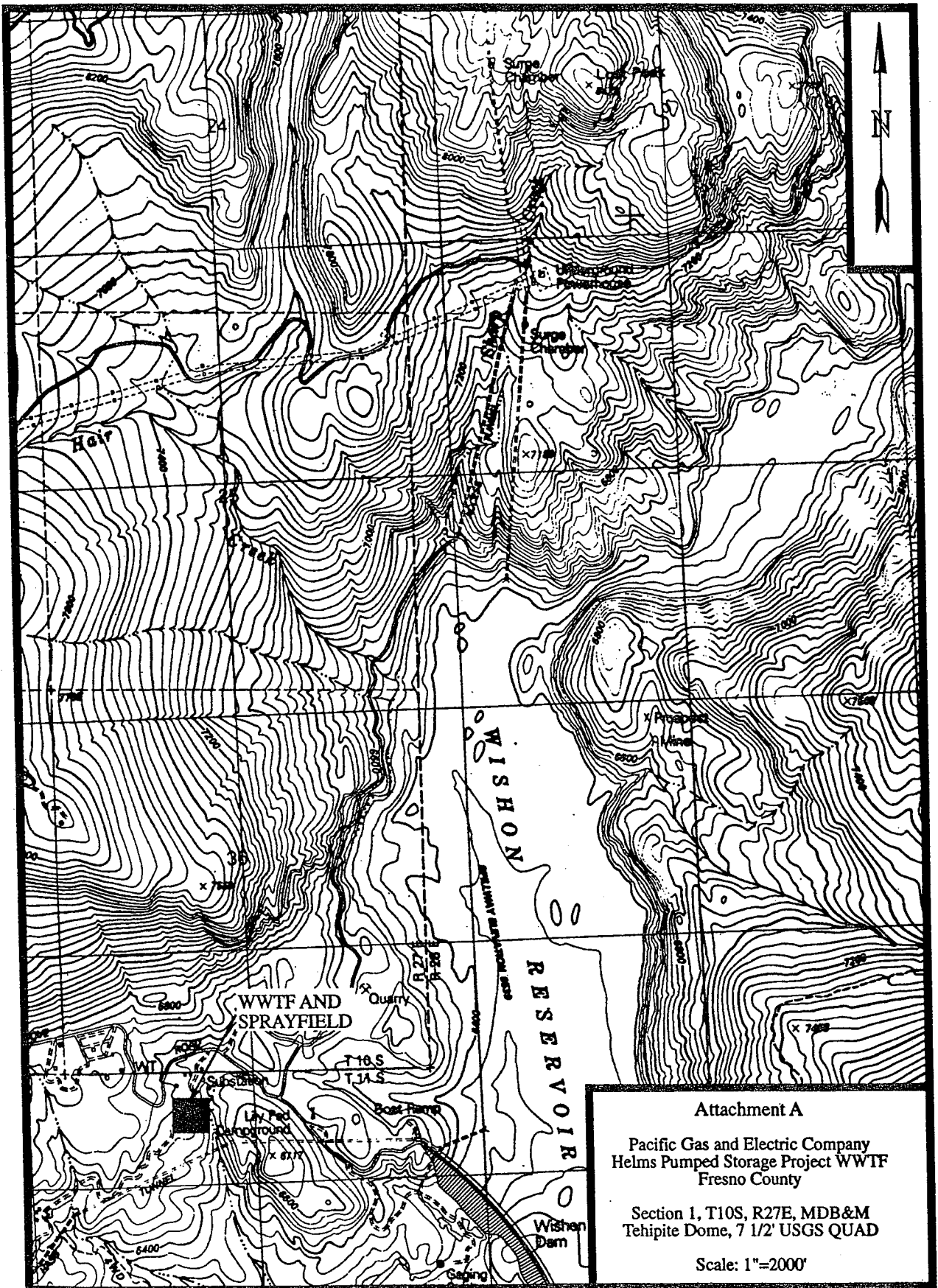
All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

Ordered by: William H. Crooks
WILLIAM H. CROOKS, Executive Officer

3 May 1996
(Date)

JLA:jla/fmc



INFORMATION SHEET

PACIFIC GAS AND ELECTRIC COMPANY HELMS PUMPED STORAGE PROJECT WASTEWATER TREATMENT FACILITY FRESNO COUNTY

Pacific Gas and Electric Company (Discharger) discharges domestic wastes from a wastewater treatment facility (WWTF) to a sprayfield for final disposal near Wishon Reservoir. The WWTF serves the Helms Pumped Storage Project headquarters and company cabins near the headquarters at Wishon and Courtright reservoirs on the Kings River drainage. The discharge to the sprayfield is presently governed by Waste Discharge Requirements Order No. 96-017.

The WWTF consists of an extended aeration treatment package plant with chlorination facilities and a sprayfield for effluent disposal. Order No. 96-107 prescribes requirements for a discharge of up to 15,000 gallons per day (gpd), but the Discharger recently verified that the actual treatment capacity of the package plant is 7,500 gpd. Wastewater flows from the cabins are 2,000 to 4,000 gpd. Until recently, the WWTF also served a privately owned campground whose wastewater contribution, coupled with inflow/infiltration, resulted in influent WWTF flows that consistently exceeded the 15,000 gpd rated plant capacity during the winter months. The Discharger disconnected the campground from its WWTF in 1995 and is mitigating I/I to reduce influent flows. The campground has its own septic tank/leachfield system available. In June 1995, the plant was also organically overloaded because solids from unit processes had not been removed periodically. This proposed Order prescribes an effluent flow limit of 7,500 gpd, which is the actual plant treatment capacity, and requires the Discharger to disinfect the effluent prior to discharge to the sprayfield.

Average annual precipitation for the site is about 50 inches, ninety percent of which is snow. Areal ground water is a few feet below ground surface and of excellent quality. The sprayfield is surrounded by surface water drainage channels and on a hillside with slopes in excess of 40 percent. Soils in the area of the sprayfield are granite and decomposed granite, topped with a few inches of humus. From late October to early May, the sprayfield is not easily accessible because it is covered with snow. Snowmelt enters the Kings River and other ephemeral streams that drain to the Valley floor.

The action to update waste discharge requirements, including this minor facility modification, is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.

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